

Java2: Lesson 9 – Lab Project

	1. Add summary at the top, documentation at the bottom and comments where necessary. - Recordings and interviews are an exception, but they should be included in your submission code. 2. Cite the source of your information. 3. Refrain from including anything you don't fully understand.		
2	50 points Write code to produce Fibonacci Numbers. 1. Implement a <i>for-loop</i> (<i>J209_21.java</i>) and recursion (<i>J209_22.java</i>). 2. Use a zero-based index. For example, to generate the first 10 numbers, the input should be 9. <i>*Do not duplicate the textbook's algorithm verbatim. Ensure your implementation has its own unique modifications or improvements. https://liveexample.pearsoncmq.com/html/ComputeFibonacci.html</i> <table> <tr> <td>File(s) to submit</td><td>J209_21.java J209_22.java</td></tr> </table>	File(s) to submit	J209_21.java J209_22.java
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3	50 points Write a method that moves the "Tower of Hanoi" using three inputs: 1. The number of disks you want to move. 2. The starting rod, which can be labeled as rod A. 3. The destination rod, which can be either rod B or rod C. 4. Test your algorithm by running it with disk values of 3, 4, and 5. <i>*Note that there are only three rods available: A, B, and C.</i> <i>*Learn the rule of the game and play it online at:</i> https://www.mathsisfun.com/games/towerofhanoi.html <i>*Here are some resources to help you understand the game and its mechanics:</i> https://liveexample.pearsoncmg.com/html/TowerOfHanoi.html https://www.geeksforgeeks.org/c-program-for-tower-of-hanoi/ https://www.digitalocean.com/community/tutorials/tower-of-hanoi https://www.mathsisfun.com/games/towerofhanoi.html <table> <tr> <td>File(s) to submit</td><td>J209_3.java J209_3.png</td></tr> </table>	File(s) to submit	J209_3.java J209_3.png
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4	50 points Update your binary search code to search for a value in the array {10, 9, 8, 7, 6, 5, 4, 3, 2, 1, 0} using a recursive method. 1. Annotate the code with proper comments. use the <i>/**</i> symbol to highlight your modifications. 2. Provide a clear and detailed, step-by-step explanation of your binary search algorithm. 3. Explain your algorithm using debugging mode in a video on YouTube. <i>* Do not duplicate the textbook's algorithm verbatim. Ensure your implementation has its own unique modifications or improvements. Avoid utilizing built-in methods like 'binarySearch()' for the core algorithm</i> <table> <tr> <td>File(s) to submit</td><td>J209_4.java J209_4.png</td></tr> </table>	File(s) to submit	J209_4.java J209_4.png
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	YouTube link																						
5	50 points Update your Palindrome code using a recursive method. 1. Add proper comments to the code, using the <code>/**</code> symbol to indicate your modifications. 2. Provide a clear and detailed, step-by-step explanation of your binary search algorithm. 3. Explain your algorithm using debugging mode in a video on YouTube. <i>* Do not duplicate the textbook's algorithm verbatim. Ensure your implementation has its own unique modifications or improvements. Avoid utilizing built-in methods like 'equals()' for the core algorithm</i> <table><tr><td>File(s) to submit</td><td>J209_4.java J209_4.png YouTube link</td></tr></table>	File(s) to submit	J209_4.java J209_4.png YouTube link																				
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6	100 points Draw a fractal of your choice from below, or create your own fractal: 1. Use Scene Builder 2. Detail your algorithm using debugging mode in IntelliJ → YouTube <table><tr><td>Apollonian gasket</td><td>Barnsley fern</td><td>Burning ship fractal</td><td>Cantor set</td></tr><tr><td>Dragon curve</td><td>Fractal tree</td><td>Hilbert curve</td><td>Hénon map</td></tr><tr><td>Julia set</td><td>Koch snowflake</td><td>Levy C curve</td><td>Lorenz attractor</td></tr><tr><td>Menger sponge</td><td>Mandelbrot set</td><td>Newton fractal</td><td>Peano curve</td></tr><tr><td>Pythagoras tree</td><td>Sierpinski triangle</td><td>T-Square fractal</td><td>Vicsek fractal</td></tr></table> <i>*For fractal images: https://youtu.be/cHJ3pjEC8Po</i> <i>*Refer to the links below for examples and resources:</i> https://liveexample.pearsoncmg.com/html/SierpinskiTriangle.html https://www.youtube.com/watch?v=0jjeOYMjmDU https://www.youtube.com/watch?v=X8bXDKqMsXE https://youtu.be/_ABdBoW4DV8 https://youtu.be/WFtTdf3l6Ug <table><tr><td>File(s) to submit</td><td>J209_5.java J209_5.fxml J209_5_controller.java J209_5.png YouTube link</td></tr></table>	Apollonian gasket	Barnsley fern	Burning ship fractal	Cantor set	Dragon curve	Fractal tree	Hilbert curve	Hénon map	Julia set	Koch snowflake	Levy C curve	Lorenz attractor	Menger sponge	Mandelbrot set	Newton fractal	Peano curve	Pythagoras tree	Sierpinski triangle	T-Square fractal	Vicsek fractal	File(s) to submit	J209_5.java J209_5.fxml J209_5_controller.java J209_5.png YouTube link
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